A Hidden Limit of Space and Time Could Lead to Ultrafast Electronics

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Quantum physics has a rule: you cannot know everything about a particle at once. Werner Heisenberg's **uncertainty** principle says a particle's position and **momentum** cannot both be measured exactly. But position and time are different. Now scientists have shown another limit: an electron's location and its behaviour over time cannot both be pinned down exactly. This new 'space-time limit' could matter for quantum technology and ultrafast electronics. The discovery comes from a team in Regensburg, Germany, working with colleagues in Hamburg. To watch **electrons** move, the team built a new laser system that works like a high-speed camera. Timed laser pulses drive **electrons** from a sharp metal tip toward a silver surface. The movements happen over **attoseconds**: one attosecond is one-billionth of a billionth of a second. At these distances, **electrons** move roughly a thousand times faster than atoms. Instead of moving like tiny balls, the **electrons** behave as quantum waves, letting them pass through an energy barrier they should not be able to cross. This is quantum **tunnelling**, a bit like passing through a solid wall without breaking it.

Word Treasure

uncertainty -- Not being able to know something exactly or for sure.
momentum -- How much motion an object has, from its mass and its speed.
attoseconds -- Extremely short periods of time, each one-billionth of a billionth of a second.
tunnelling -- When a tiny particle passes through a wall it should not be able to cross.
electrons -- Tiny particles that move around the centre of an atom.

Background you need

Quantum physics is the science of the very smallest things, like electrons.

An electron is a tiny particle that moves around the centre of an atom.

A laser makes a narrow beam of light, and short laser pulses can work like a super-fast camera.

Break it down

WHO: An international team of scientists in Regensburg and Hamburg.
WHAT: Scientists measured a new 'space-time limit' for electrons.
WHERE: In a lab in Germany, using a new laser system.
WHY: To watch electrons move and learn the speed limits of future electronics.

Why it matters

If you use a phone or a computer, this science could one day help make them much, much faster.

Test what you remember

1. What does the uncertainty principle say?

- (A) A particle's position and momentum cannot both be measured exactly
- (B) Electrons always move in straight lines
- (C) Light is the fastest thing
- (D) Time travel is possible

2. What is the new limit about?

- (A) How cold space can get
- (B) An electron's location and its behaviour over time
- (C) How fast a rocket can fly
- (D) How bright a laser can be

3. Where is the research team?

- (A) Tokyo and Beijing
- (B) Regensburg, Germany, and Hamburg
- (C) London and Paris
- (D) New York and Boston

4. How long is an attosecond?

- (A) One-thousandth of a second
- (B) One second
- (C) One-billionth of a billionth of a second
- (D) One-millionth of a second

5. How do the electrons behave instead of like tiny balls?

- (A) As quantum waves
- (B) As tiny magnets
- (C) As beams of light
- (D) As drops of water

6. What is quantum tunnelling like?

- (A) Passing through a solid wall without breaking it
- (B) Digging a tunnel through a mountain
- (C) Sending a message through space
- (D) Heating metal until it melts

Answer key (try the quiz first!): 1: A 2: B 3: B 4: C 5: A 6: A

Questions to think about

1. Lead author Simon Maier: He says that by changing the time between the two laser pulses, the team can directly observe how the electrons respond.

2. Co-author Katharina Glöckl: She says the measurement can be understood as a high-speed camera for the electron wave packets.

MY THOUGHTS (at least 20 words):